

Prerequisites

- ⚠ Important
- Use Kiro CLI for this section
 - Complete [Setting Up Kiro](#)

Situation 1: Building Centralized CloudWatch Dashboard

Let's use Kiro CLI to help us

1. In your VS Code Server environment, create a new folder to separate context from other files inside VS Code Server:

```
1 cd /Workshop
2 mkdir cloudwatch-dashboard
3 cd cloudwatch-dashboard
```

2. Within the VS Code Server environment, start Kiro CLI by running:

```
1 kiro-cli chat
```

3. Ask Kiro CLI to create a dashboard for us

```
1 Create a CloudWatch monitoring dashboard using AWS best practices for the following workload
2 CloudFormation name: sample-app
3 Region: us-east-1
4 Before creating the dashboard, try querying the metrics first to ensure they exist, then create the dashboard
```

```
● Completed in 0.0s

> ## Summary

I have successfully created comprehensive CloudWatch monitoring dashboards for your sample-app workload following AWS best practices. Here's what was accomplished:

### ✅ Metrics Verification
• Verified all metrics exist before creating dashboards
• Confirmed ALB, EC2, DynamoDB, and Auto Scaling metrics are available
• Identified active instances and resources

### 📊 Dashboards Created
1. SampleApp-Monitoring-Dashboard - Basic monitoring with essential metrics
2. SampleApp-Enhanced-Monitoring - Advanced dashboard with KPIs, single-value widgets, and detailed visualizations

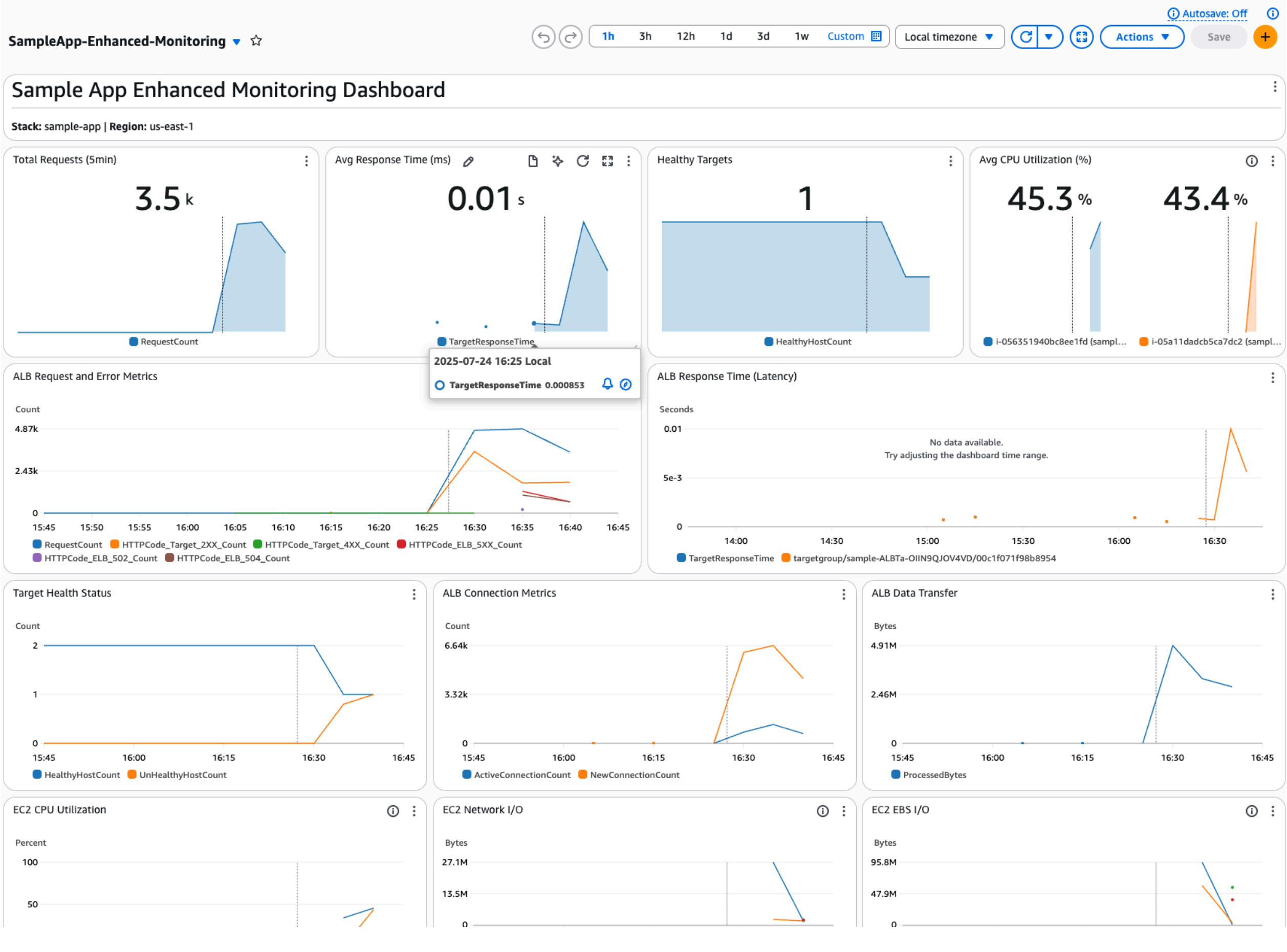
### 🚨 CloudWatch Alarms Configured
• **ALB High Response Time** (threshold: 1 second)
• **ALB Unhealthy Hosts** (currently in ALARM - needs attention)
• **EC2 High CPU** (threshold: 80%)
• **DynamoDB Throttling** (currently in ALARM - needs attention)

### 📈 Monitoring Coverage
• **Application Load Balancer**: Request metrics, response times, health status, connections, data transfer
• **EC2 Instances**: CPU utilization, network I/O, EBS I/O across both active instances
• **DynamoDB**: Capacity utilization, performance metrics, throttling detection
• **Cost Optimization**: ALB LCU monitoring for cost management

### ⚠ Current Issues Detected
• **Unhealthy ALB targets** - requires investigation
• **DynamoDB throttling** - capacity needs adjustment

### 🏠 Best Practices Implemented
• Golden signals monitoring (latency, traffic, errors, saturation)
• Proactive alerting with appropriate thresholds
• Visual hierarchy with KPIs and detailed metrics
• Cost optimization visibility
• Well-Architected Framework observability principles
```

4. Navigate to the [CloudWatch Dashboard](#) to see the dashboard



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